

LU SHEN

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Center for Astronomy and Astrophysics, Physics Department, Fudan University
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RESEARCH INTERESTS:

- My research has particularly focused on galaxy evolution at intermediate-to-high redshift, including resolved star formation, galaxy morphology, environmental effect, the coevolution of star formation and AGN in galaxies, and their dust properties.

WORK

Assistant professor Texas A & M University, College Station, Texas, United States;	April 2026 - now
Postdoctoral researcher Texas A & M University, College Station, Texas, United States;	September 2022 - April 2026
Associate researcher University of Science and Technology of China, Hefei, Anhui, China;	August 2021 - August 2022
Postdoctoral researcher University of Science and Technology of China, Hefei, Anhui, China;	September 2019 - August 2021

EDUCATION

Ph.D., Physics University of California, Davis; Davis, CA Advisor: Prof. Lori Lubin	September 2014 - August 2019
M.S., Physics University of California, Davis; Davis, CA	September 2014 - December 2015
B.S., Physics Xiamen University; Xiamen, CHINA Graduated with First Class	September 2010 - June 2014
Exchange Student University of California, Santa Barbara; Goleta, CA	September 2012 - June 2013

GRANTS

NASA JWST program ID 5075 "Unveiling the Morphological Evolution of Galaxies in Protoclusters: Insights from JWST Imaging"	2024 – 2026 \$193k (PI)
National Science Foundation for Young Scientists of China	2020 – 2022

INVITED TALKS

University of California, Davis <i>Seminar</i>	Mar 6 2025
Tsinghua University <i>Seminar</i>	Jun 4 2024
Shanghai Observatory <i>Seminar</i>	Dec. 13 2021
Nanjing University <i>Seminar</i>	Dec. 7 2021
Harvard CfA <i>Galaxy Cluster Group Meeting</i>	Mar. 9 2021
Chinese Academy of Sciences South America Center for Astronomy 2020 <i>Seminar</i>	Dec 22
Nanjing University <i>Colloquium</i>	Sept. 27 2020
KIAA-Peking University <i>Colloquium</i>	Jun. 19 2019
University of Science and Technology of China <i>Colloquium</i>	Jun 18. 2019

CONFERENCES AND WORKSHOPS

Santa Cruz Galaxy Workshop <i>Contributed talk</i>	Aug 2025 Santa Cruz, US
CFC2025: The Inaugural Cosmic Frontier Center Conference <i>Contributed talk</i>	May 2025 Austin, US
KITP program: Cosmic Origins: The First Billion Years <i>Contributed talk</i>	Aug 2024 Santa Barbara, US
2024 STScI Spring Symposium Workshops <i>Contributed talk</i>	April 15 - 19, 2024 Baltimore, US
CEERS Team Meeting <i>Contributed talk</i>	May 13 - 15 2024 San Lorenzo de El Escorial, Spain
European Astronomical Society Annual Meeting <i>Two contributed talks</i>	Jul. 10 - 14 2023 Kraków, Poland
CEERS Team Meeting <i>Contributed talk</i>	May 13 - 15 2023 Austin, US
Galaxy Cluster Formation II Workshop <i>Poster</i>	Jun. 14 - 18 2021 Virtual
Galaxies: Star formation and nuclei activity <i>Contributed talk</i>	Dec. 2 - 6 2020 Suzhou, China
Protoclusters: galaxies in confinement <i>Poster</i>	Aug. 31 - 4, 2020 Virtual

Chinese Astronomical Society Meeting <i>Contributed talk</i>	Oct. 14 2020 <i>Virtual</i>
Galaxies: Star formation and nuclei activity <i>Contributed talk</i>	Nov.26 - 28 2019 <i>Nanjing, China</i>
2019 TMT Science Forum <i>Contributed talk</i>	Nov. 4 - 6, 2019 <i>Xiamen, China</i>
2018 TMT Science Forum <i>Poster</i>	Dec. 10 - 12 2018 <i>Pasadena, USA</i>
2018 TMT Early-Career Workshop <i>Participate</i>	Dec. 2 - Dec. 9 2018 <i>Pasadena, USA</i>
2017 TMT Future Leaders Workshop <i>Participate</i>	Aug. 21 - Aug. 30 2017 <i>Santa Cruz, USA</i>
2017 ISEE Professional Development Program <i>Participate</i>	Mar. 2017 - Aug. 2017 <i>Santa Cruz, USA</i>
Preparing TMT Future Leaders <i>Participate</i>	Dec. 3 - Dec. 7 2016 <i>Hilo, USA</i>
15th Synthesis Imaging Workshop <i>Participate</i>	Jun. 1 - Jun. 8 2016 <i>New Mexico, USA</i>
Rudolph Minkowski Observational Astronomy Workshop <i>Participate</i>	Oct. 15 - Oct. 19 2015 <i>Lick Observatory, USA</i>

OBSERVATION

- MOSFIRE/Keck, DEIMOS/Keck
- MOIRCS/Subaru telescope
- CWI/Hale Telescope at Palomar Observatory

SKILL

- Most experienced in Python
- Proficient in CASA and AIPS for interferometry data reduction and analysis
- Proficient in GALFIT and statmorph for galaxy morphology measurements
- Working knowledge of UNIX, IDL, C++, GitHub

TEACHING AND OUTREACH

Teaching Assistant	Sept. 2014 - Sept 2017
· AST 10G - Introduction to Stars, Galaxies, and the Universe · AST 10S - Astronomy of the Solar System · PHYS 10G - Introduction to Stars, Galaxies, black holes, space time, and relativity · PHYS 219A - Statistical Mechanics · 9B Lab - Classical Physics	

BIBLIOGRAPHY

The ADS library link for all Lu Shen's publications.

First Author:

NGDEEP: The Star Formation and Ionization Properties of Galaxies at $1.7 < z < 3.4$
submitted

Shen, L; Papovich, C; Matharu, J; Pirzkal, N; Hu, W; and the NGDEEP Collaboration

NGDEEP Epoch 1: Spatially Resolved H α Observations of Disk and Bulge Growth in Star-forming Galaxies at $z \sim 0.6-2.2$ from JWST NIRISS Slitless Spectroscopy

2024, ApJ, 963L, 49S

Shen, L; Papovich, C; Matharu, J; Pirzkal, N; Hu, W; and the NGDEEP Collaboration

CEERS: Spatially Resolved UV and mid-IR Star Formation in Galaxies at $0.2 < z < 2.5$: The Picture from the Hubble and James Webb Space Telescopes

2023, ApJ, 950, 7S

Shen, L; Papovich, C; Yang, G; Matharu, J; Wang, X; Magnelli, B; Elbaz, D; Jogee, S, the CEERS Collaboration and UVCANDELS Collaboration

Discovery of spectacular quasar-driven superbubbles in red quasars

2023, Science Advance, vol. 9, issue 28

Shen, L; Liu, G.; He, Z.; Zakamska, N. L. ; Glikman, E. Greene, J. E.; Hu, W.; Mou, G.; Wylezalek, D.; Rupke, D. S. N. ;

The ALPINE-ALMA [CII] survey: The infrared-radio correlation and AGN fraction of star-forming galaxies at $z \sim 4.4 - 5.9$

2022, ApJ, 935, 177S

Shen, L, Lemaux, B. C, Lubin, L. M., Liu, G, Béthermin, M., Boquien, M., Cucciati, O, Le Fèvre, O., Talia, M., Vergani, D., Zamorani, G, Faisst, A. L., Ginolfi, M., Gruppioni, C., Jones, G., Bardelli, S., Hathi, N., Koekemoer, A. M., Romano, M., Zucca, E., Fang, W., Forrest, B., Gal, G., Hung, D., Shah, E. A., Staab, P., Vanderhoof, B.

Implications of the Environments of Radio-detected AGN in a Complex Proto-structure at $z \sim 3.3$

2021, ApJ, 912, 60S

Shen, L, Lemaux, B. C, Lubin, L. M., Cucciati, O, Le Fèvre, O, Liu, G, Fang, W, Pelliccia, D, Tomczak, A, McKean, J, Miller, N. A, Fassnacht, C. D. D, Gal, R, Hung, Hathi, N, Bardelli, S, Vergani, D, Zucca E

Extended Radio AGN at $z \sim 1$ in the ORELSE survey: The confining effect of dense environments

2020, ApJ, 902, 101S

Shen, L, Liu, G, Zhang, M, Lemaux, B. C, Lubin, L. M., Pelliccia, D, Moravec, E, Golden-Marx, E, Zhou, H, Fang, W, Tomczak, A, McKean, J, Miller, N. A, Fassnacht, C. D., Wu, P, Kocevski, D, Gal, R, Hung, D, Squires, G. K.

The Properties of Radio and Mid-infrared Detected Galaxies and the Effect of Environment on the Co-evolution of AGN and Star Formation at $z \sim 1$

2020, MNRAS, 494, 5374

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Shen, L., Lemaux, B. C., Lubin, L. M., McKean J., Miller. N. A., Pelliccia, D., Fassnacht, C. D., Tomczak, A. R., Wu, P.-F., Kocevski, D., Gal, R. R., Hung, D. and Squires, G. K.

Possible Evidence of the Radio AGN Quenching of Neighboring Galaxies at $z \sim 1$

2019, MNRAS, 484, 2433S

Shen, L., Tomczak, A. R., Lemaux, B. C., Pelliccia, D., Lubin, L. M., Miller. N. A., Fassnacht, C. D., Becker, R. H., Gal, R. R., Wu, P.-F., Squires, G. K.

The Properties of Radio Galaxies and the Effect of Environment in Large Scale Structures at $z \sim 1$

2017, MNRAS, 472, 998

Shen, L., Miller, N. A., Lemaux, B. C., Tomczak, A. R., Lubin, L. M., Rumbaugh, N., Fassnacht, C. D., Becker, R. H., Gal, R. R., Wu, P.-F., and Squires, G. K.

Coauthor:

CEERS: 7.7 μ m PAH Star Formation Rate Calibration with JWST MIRI

Ronayne et al. 2023, arXiv:2310.07766

CEERS: MIRI deciphers the spatial distribution of dust-obscured star formation in galaxies at $0.1 < z < 2.5$

Magnelli et al. 2023, A&A, 678A, 83M

CEERS Key Paper. V. Galaxies at $4 < z < 9$ Are Bluer than They Appear-Characterizing Galaxy Stellar Populations from Rest-frame 1 μ m Imaging

Papovich et al. 2023, ApJ, 949L, 18P

CEERS Key Paper. VI. JWST/MIRI Uncovers a Large Population of Obscured AGN at High Redshifts

Yang et al. 2023, ApJ, 950L, 5Y

A Long Time Ago in a Galaxy Far, Far Away: A Candidate $z \sim 12$ Galaxy in Early JWST CEERS Imaging

Finkelstein et al. 2022, ApJL, 940L, 55F

Evidence for quasar fast outflows being accelerated at the scale of tens of parsecs

He, Z et al. 2022, SciA, 8, 3291H

The VIMOS Ultra Deep Survey: The Reversal of the Star Formation Rate – Density Relation at $2 < z < 5$

Lemaux, B. C. et al. 2022, A&A, 662A, 33L

B2 0003+38A: A Classical Flat-spectrum Radio Quasar Hosted by a Rotation-dominated Galaxy with a Peculiar Massive Outflow

Zhao, Q. et al. 2021, ApJ, 913, 111Z

An optical observational cluster mass function at $z \sim 1$ with the ORELSE survey

Hung, D. et al. 2021, MNRAS, 502, 3942H

The High-redshift Clusters Occupied by Bent Radio AGN (COBRA) Survey: Radio Source Properties

Golden-Marx et al. 2021, ApJ, 907, 65G

Effects of Stellar Feedback on Stellar and Gas Kinematics of Star-forming Galaxies at $0.6 < z < 1.0$

Pelliccia, D. et al. 2020, ApJ, 896L, 26P

Establishing a New Technique for Discovering Large-Scale Structure Using the ORELSE Survey

Hung, D. et al. 2020, MNRAS, 491, 5524

Persistence of the Color-Density Relation and Efficient Environmental Quenching to $z \sim 1.4$

Lemaux, B. C et al. 2019, MNRAS, 490, 1231

Conditional Quenching: A detailed look at the SFR-Density Relation at $z \sim 0.9$ from ORELSE

Tomczak, A. R. et al. 2019, MNRAS, 484, 4695

Searching for environmental effects on galaxy kinematics in groups and clusters at $z \sim 1$ from the ORELSE survey

Pelliccia, D. et al. 2019, MNRAS, 482, 3514

Evaluating Tests of Virialization and Substructure Using Galaxy Clusters in the ORELSE Survey

Rumbaugh, N. et al. 2018, MNRAS, 478, 1403

Similar Scaling Relations for the Gas Content of Galaxies Across Environments to $z \sim 3.5$

Darvish, B. et al. 2018, ApJ, 860, 111D

Glimpsing the Imprint of Local Environment on the Galaxy Stellar Mass Function

Tomczak, A. R. et al. 2017, MNRAS, 472, 3512